N. Robertson and P. D. Seymour, Graph Minors. III. Planar Tree-Width, *J. Combin. Theory Ser. B* **36** (1984), 49–64.

About this copy

This is a corrected copy of the paper cited above. The text is retyped from the printed paper. It keeps the paper's wording and the numbering of its results, equations and footnotes, and a marker such as [p. N] shows where page N of the printed paper begins.

The corrections come from a formalization of the Graph Minors series in the Lean proof assistant, github.com/mahdi-jfri/GraphMinor, in which the results of this paper are proved. Where the formalization found that a statement or a step of a proof is wrong as printed, this copy corrects it in place. Where the paper leaves a proof or a step to the reader, this copy adds it, also in blue. Text ~~struck through in red~~ is removed and text in blue is inserted. Each change has a note, marked F1, F2, ..., that gives the reason. The file corrections/GM03.md of the formalization describes each correction in full. Each note, and each item in the list below, begins with the kind of the change. A **misprint** is a wrong symbol, index or word. A **statement mistake** is a result or a claim of the text that is false as printed. A **degenerate case** is a statement that fails only in a trivial case, such as an empty graph or a parameter equal to 0. A **proof gap** is a step of a proof that does not follow as written. A **missing proof** is a proof that the paper omits, leaves to the reader or only sketches, and that this copy adds.

The corrections and added proofs, in the order of the pages:

1. **Missing proof.** Proof of (2.2), Claim (2), pages 53–54: the verification that the enlarged graph satisfies (i)–(iv), including the radius, is added.
2. **Missing proof.** Proof of (2.3), page 55: the omitted part is added.
3. **Misprint.** Proof of (2.5), page 55: in the definition of t', "if t' is not the center" becomes "if t is not the center".
4. **Missing proof.** Proof of (2.5), pages 55–56: the verification of the tree-decomposition is added.
5. **Missing proof.** Proof of (2.6), page 56: the proof is added.
6. **Missing proof.** Proof of (2.8), page 56: the proof is added.
7. **Degenerate case.** Section 3, page 59: in the definition of a sleeve union, s is the least *positive* integer with $2s \geq r$, so that the cylinder exists also for $r = 0$.

[p. 49] Graph Minors. III. Planar Tree-Width

Neil Robertson and P. D. Seymour*
Department of Mathematics, Ohio State University, 231 West 18th Avenue, Columbus, Ohio 43210
Communicated by the Editors
Received January 14, 1983

The "tree-width" of a graph is defined and it is proved that for any fixed planar graph H, every planar graph with sufficiently large tree-width has a minor isomorphic to H. This result has several applications which are described in other papers in this series.

1. Introduction

All graphs in this paper are finite, and may have loops or multiple edges. $V(G)$ and $E(G)$ denote the sets of vertices and edges of the graph G, respectively. A *tree-decomposition* of a graph G is a pair $(T, \mathcal{X})$, where T is a tree and $\mathcal{X} = (X_t : t \in V(T))$ is a family of subsets of $V(G)$, with the following properties:

(W1) $\bigcup_t (X_t : t \in V(T)) = V(G)$.

(W2) For every edge e of G there exists $t \in V(T)$ such that e has both ends in X_t.

(W3) For $t, t', t'' \in V(T)$, if t' is on the path of T between t and t'' then

$$X_t \cap X_{t''} \subseteq X_{t'}.$$

The width of the tree-decomposition is

$$\max_{t \in V(T)} (|X_t| - 1).$$

The graph G has *tree-width* w if w is minimum such that G has a tree-decomposition of width w.

Thus, for example, trees and forests have tree-width 1 (or 0 in degenerate [p. 50] cases) and series-parallel graphs have tree-width ≤ 2. For $n \geq 1$, the complete graph K_n has tree-width $n - 1$, while for $n \geq 2$ the $n \times n$ grid (this has n^2 vertices—it is the adjacency graph of the chessboard with n^2 squares) can be shown to have tree-width n.

*Research partially supported by NSF Grant MCS 8103440.

A graph H is a *minor* of a graph G if H can be obtained from a subgraph of G by contraction. If $H \neq G$ it is a *proper minor*. Other papers in this series prove theorems concerning minors of graphs chosen from a set of graphs with bounded tree-width. For example,

(1.1) [3]. *If $\mathcal{F}$ is an infinite set of graphs all with tree-width $\leq w$, where w is some integer, then there exist $G, G' \in \mathcal{F}$ such that G is isomorphic to a minor of G'.*

(1.2) [2]. *If $\mathcal{F}$ is a (possibly infinite) set of graphs, all with tree-width $\leq w$, where w is some integer, then there is a polynomial algorithm which, given an arbitrary graph G, tests if there is a graph $G' \in \mathcal{F}$ such that G is isomorphic to a minor of G'.*

The purpose of this paper is to provide a useful source of sets of graphs with bounded tree-width. The set of all planar graphs is not such a set, because it contains all the grids. However, our main result may be expressed as follows.

(1.3) *Any proper subset of the set of all planar graphs which is closed under isomorphism and the taking of minors is a set of graphs with bounded tree-width.*

This may be reformulated in the following more convenient form.

(1.4) *For every planar graph H, there is a number w such that every planar graph with no minor isomorphic to H has tree-width $\leq w$.*

(1.1) and (1.4) together imply that planar graphs are "well-quasi-ordered" by the taking of minors; more precisely,

(1.5) *If $\mathcal{F}$ is an infinite set of planar graphs, then there exist distinct $G, G' \in \mathcal{F}$ such that G is isomorphic to a minor of G'.*

Proof. Choose $H \in \mathcal{F}$, and choose a number w as in (1.4). If there exists $G \in \mathcal{F} - \{H\}$ such that H is isomorphic to a minor of G then (1.5) is true. If not, then by our choice of w, each $G \in \mathcal{F} - \{H\}$ has tree-width $\leq w$, and the result follows from (1.1).

We remark that in [4] we shall prove that (1.4) is true with the second occurrence of "planar" deleted. We are publishing the weaker version of

[p. 51] (1.4) separately here because it is much easier to prove than the result of [4], and our proof here yields a very much better estimate of the value of w than that in [4].

The proof of (1.4) occupies the whole paper (except for some remarks at the end) and we begin by sketching it. It is enough to prove (1.4) when H is a "cylinder" (a grid with one pair of opposite sides identified) because every

planar graph is isomorphic to a minor of some sufficiently large cylinder. We introduce the "sleeve union" of two graphs G_1, G_2, and observe that the tree-width of the resulting graph G is the maximum of the tree-widths of G_1, G_2, and that G_1, G_2 are isomorphic to minors of G. It follows that it is enough to prove that if a planar graph is not expressible as a sleeve union of two smaller planar graphs, and has no minor isomorphic to the $n \times n$ cylinder, then its tree-width is bounded by a function of n. We prove that in any plane drawing of such a graph there do not exist $n + \frac{1}{2}n^2$ vertex-disjoint circuits each inside the next, and deduce that it has tree-width $\leq 3(n + \frac{1}{2}n^2) - 2$. These steps are performed more or less in reverse order.

Let us clarify some terminology. A *walk* in a graph G is a sequence $v_1, e_1, v_2, e_2, \ldots, v_k, e_k, v_{k+1}$ of vertices and edges of G (where $k \geq 0$), such that for $1 \leq j \leq k$, the ends of e_j are v_j and v_{j+1}. The walk is closed if $v_1 = v_{k+1}$. A *path* is a walk such that $v_j \neq v_{j'}$ for $1 \leq j < j' \leq k+1$, and a *circuit* is a closed walk with $k \geq 1$ such that $v_j \neq v_{j'}$ for $1 \leq j < j' \leq k$. We often identify paths and circuits with the corresponding subgraphs of G when no confusion should arise. The *initial* and *terminal* vertices of the walk are v_1, v_{k+1}. When $u, v \in V(G)$, a path between u and v is a path with initial vertex u and terminal vertex v, and when $X, Y \subseteq V(G)$, a path between X and Y is a path with initial vertex in X and terminal vertex in Y. A path *avoids* $Z \subseteq V(G)$ if no vertex of it is in Z. When $Z \subseteq V(G)$, Z *separates* X and Y if no path of G between X and Y avoids Z. (Here X, Y may be vertices or sets of vertices.) If G_1, G_2 are subgraphs of G we denote by $G_1 \cup G_2$ the subgraph of G consisting of the vertices in $V(G_1) \cup V(G_2)$ and the edges in $E(G_1) \cup E(G_2)$. We define $G_1 \cap G_2$ similarly. If X is a vertex or an edge of G, or a set of vertices or edges, we denote by $G \setminus X$ the graph obtained by deleting X from G.

2. Radius

If G is a planar graph, it can be drawn without crossings in the plane, and by a "drawing" we shall always mean such a drawing. If M is a drawing of G and C is a circuit of G, then every region of M and every vertex and edge of G not in C is either inside or outside C in M, in the obvious sense. For every region R of a drawing M, we define $d(R)$ to be the minimum value of k such that there is a sequence $R_0, R_1, \ldots, R_k$ of regions of M, where R_0 is the [p. 52] infinite region, $R_k = R$, and for $1 \leq j \leq k$ there is a vertex v incident with both R_{j-1} and R_j. The *radius* $\rho(M)$ of M is the minimum value of d such that $d(R) \leq d$ for all regions R of M. The *radius* of a planar graph is the minimum of the radii of its drawings.

The object of this section is to prove that if a planar graph has radius d then its tree-width is at most $3d + 1$. The proof takes several steps.

(2.1) *Any planar graph of radius $\leq d$ is isomorphic to a minor of a simple planar graph with radius $\leq d$.*

Proof. We replace each edge of the graph (G, say) by three edges in series, forming a new graph G'. It is clear that G' is planar and has the same radius as G, and that G' is simple. Moreover G is isomorphic to a minor of G', as required.

Let M be a drawing of a planar graph G. By a *nested sequence of circuits* of M we mean a sequence $C_1, \ldots, C_d$ of circuits of G such that for $1 \leq j < j' \leq d$, C_j and $C_{j'}$ have no common vertices and $C_{j'}$ is inside C_j in M. By a *d-shell* of M (where $d \geq 0$ is an integer) we mean a nested sequence $C_1, \ldots, C_d$ of circuits of M such that

(S1) every vertex of G is either in one of $V(C_1), \ldots, V(C_d)$ or is inside C_d,

(S2) for $1 \leq j \leq d$, every edge of G which joins two vertices in $V(C_j)$ is in $E(C_j)$.

(2.2) *Let G be a simple planar graph of radius $\leq d$, where $d \geq 0$ is an integer. Then there exist a simple planar graph G' with a minor isomorphic to G and a drawing M' of G' such that $\rho(M') \leq d$ and M' has a d-shell.*

Proof. For $d' = 0, 1, \ldots, d$, we shall prove by induction on d' that there exist a simple planar graph G' with a minor isomorphic to G and a drawing M' of G' such that M' has a d'-shell and $\rho(M') \leq d$. This is clearly true when $d' = 0$. Now assume that $0 < d' \leq d$ and assume the inductive hypothesis that there is a planar graph G' with a drawing M' such that

(i) G' is simple, G is isomorphic to a minor of G', and $\rho(M') \leq d$

(ii) M' has a $(d' - 1)$-shell $C_1, \ldots, C_{d'-1}$.

Let H be the graph obtained from G' by deleting the vertices in $V(C_1), \ldots, V(C_{d'-1})$ and the edges incident with them; and let N be the drawing of H obtained from M' by the same deletion.

We may assume that

(iii) at least three vertices of H are on the infinite region of N.

[p. 53] For if not, we may add isolated vertices to G' and to M' to make (iii) true without falsifying (i) or (ii).

We may also assume that

(iv) *every vertex of H is adjacent to at most one vertex of $C_{d'-1}$.*

For we may replace every edge of G' joining $V(H)$ and $V(C_{d'-1})$ by two edges in series, thereby making (iv) true without falsifying any of (i), (ii), or (iii).

Let $|V(H)| = k$, say. We may also assume that

(v) *subject to (i)–(iv) and to* $|V(H)| = k$, G' *is chosen with* $|E(H)|$ *maximum.*

(This is possible because G' is constrained to be simple and $|V(H)|$ is fixed.)

Let $C_{d'}$ be the subgraph of H consisting of the vertices and edges incident with the infinite region of N.

(1) $C_{d'}$ *is connected.*

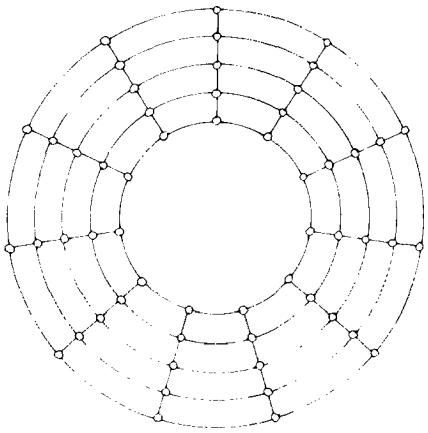
For if not, it follows from (S2) applied to $C_{d'-1}$ that there are vertices v_1, v_2 of $C_{d'}$ in different components of $C_{d'}$, and hence of H but incident with the same region of M'. We add an edge joining them. The resulting graph still satisfies (i)–(iv), contrary to (v).

(2) $C_{d'}$ *is a circuit.*

For $C_{d'}$ has at least three vertices by (iii). If it is not a circuit, it has a vertex v such that $C_{d'} \setminus v$ is disconnected, and hence such that $H \setminus v$ is disconnected. Let D be the vertex set of one component of $C_{d'} \setminus v$, and let

$$D' = V(C_{d'}) - (D \cup \{v\}).$$

Let $v_1, e_1, v_2, e_2, \ldots, v_k, e_k, v_{k+1}(= v_1)$ be the closed walk in H which is the perimeter of the infinite region of N. There exist distinct i, j with $1 \le i, j \le k$, such that $v_i = v_j = v$, and $v_{i-1}, v_{j+1} \in D$, and $v_{i+1}, v_{j-1} \in D'$ (where v_0 means v_k; similarly, later, e_{k+1} will mean e_1, and e_0 will mean e_k). Now by (iv), there is at most one edge of G' incident with v which is not in $E(H)$; and so without loss of generality we assume that there is a region R of M' for which the path $v_{i-1}, e_{i-1}, v, e_i, v_{i+1}$ forms part of the perimeter, and for which (a) if $d' = 1$, R is the infinite region of M', and (b) if $d' > 1$, R is incident with a vertex in $C_{d'-1}$. We add a new edge e to G' joining v_{i-1} and v_{i+1}, and make the corresponding extension of M'. R is thus divided into two regions, one incident with v_{i-1}, e, v_{i+1} and (a) being the infinite region if $d' = 1$, and (b) being incident with a vertex of $C_{d'-1}$ if $d' > 1$, and the other a triangle bounded by $v_{i-1}, e_{i-1}, v, e_i, v_{i+1}, e$. We claim that this enlarged graph satisfies (i)–(iv), contrary to (v). Verifying this is straightforward ~~and is left to the reader~~. (The only part which is not quite



[p. 54]

FIGURE 1

trivial is checking the radius, using the condition $d' - 1 < d$, and this too is easy.)

We give the details. Let G'' and M'' be the enlarged graph and drawing, let H'' be H with the new edge e added, and let N'' be its drawing. The edge e joins two vertices of H and lies in R, and so (ii) and (iv) still hold, and G is still isomorphic to a minor of G''. Since $v_{i-1} \in D$ and $v_{i+1} \in D'$ lie in different components of $H \setminus v$, they are distinct and not adjacent in G', and so G'' is simple. The perimeter of the infinite region of N'' is the closed walk above with $v_{i-1}, e_{i-1}, v, e_i, v_{i+1}$ replaced by v_{i-1}, e, v_{i+1}. It has the same vertices, since $v = v_j$ is still on it, and so (iii) holds. It remains to show that $\rho(M'') \leq d$. Call a region of M' *inner* if it is contained in the infinite region of N and, when $d' > 1$, it is inside $C_{d'-1}$. Call a region of M'' *inner* if the same holds with N'' in place of N. The region R is an inner region of M', and M'' differs from M' only inside R. So every region of M'' is inner, or is the triangle, or is a region of M' that is not inner.

(α) *Let $d' > 1$ and $1 \leq h \leq d' - 1$. Every vertex of C_h is incident with a region of M' outside C_h that has distance at most $h - 1$ in M', by a sequence of regions outside C_h.*

We use induction on h. By (S1) no vertex is outside C_1, and by (S2) no edge is outside C_1. So the infinite region of M' is bounded by C_1, and it is incident with every vertex of C_1. This proves the case $h = 1$. Now let $1 \leq h < d' - 1$ and $u \in V(C_{h+1})$. Some region Q of M' inside C_h and outside C_{h+1} is incident with u. By (S1) and (S2) no vertex lies strictly between C_h and C_{h+1}, and every edge strictly between them has an end in C_h. So Q is incident with a vertex of C_h, for otherwise it would be bounded

by C_{h+1} alone, and would contain C_h. By induction that vertex is incident with a region outside C_h of distance at most $h-1$, by a sequence of regions outside C_h. Adding Q to this sequence proves (α) for $h+1$.

Since M'' differs from M' only inside $C_{d'-1}$, (α) also holds for M''.

(β) *Every inner region of M'' has distance at most $d'-1$ in M''.*

If $d'=1$, then $N''=M''$, and the only inner region is the infinite region of M''. Let $d'>1$, and let Q be an inner region of M''. Then Q is incident with a vertex of $C_{d'-1}$. For otherwise, since every vertex inside $C_{d'-1}$ is in H by (S1), every edge on the boundary of Q is an edge of H''. Then Q would be a region of N'', contained in the infinite region of N'' and so equal to it. This is impossible, since that region contains $C_{d'-1}$ and Q is inside $C_{d'-1}$. By (α) with $h=d'-1$, that vertex is incident with a region of distance at most $d'-2$, and so Q has distance at most $d'-1$.

(γ) *Every vertex of H'' on the perimeter of the infinite region of N'' is incident with an inner region of M'', which has distance at most $d'-1$.*

For such a vertex x is incident with a region of M'' contained in the infinite region of N''. That region is inside $C_{d'-1}$ if $d'>1$, since x is. So it is inner, and it has distance at most $d'-1$ by (β).

(δ) *Every region Q of M'' that is a region of M' and is not inner has distance at most d in M''.*

Let $R_0, R_1, \ldots, R_k = Q$ be regions of M' as in the definition of $d(Q)$ in M', with $k \le d$. If none of them is inner, they are regions of M'', and consecutive ones share a vertex of M'', and so Q has distance at most k in M''. Otherwise let m be the largest index such that R_m is inner. Then $m<k$, and $R_{m+1}, \ldots, R_k$ are regions of M''. Moreover $m \ge d'-1$. For two regions incident with a common vertex are on the same side of every C_h that does not contain that vertex, and so two consecutive regions of the sequence are separated by at most one of $C_1, \ldots, C_{d'-1}$, while R_0 is outside C_1 and R_m is inside $C_{d'-1}$. Let w be a vertex incident with R_m and R_{m+1}. If $w \in V(H)$, then w is on the perimeter of the infinite region of N, since R_m is contained in that region, and this perimeter has the same vertices as that of N''. By (γ), w is incident with a region of M'' of distance at most $d'-1$, and so Q has distance at most $d'-1+k-m \le k \le d$. Otherwise $d'>1$ and $w \in V(C_{d'-1})$, by (S1). By (α), w is incident with a region of M'' of distance at most $d'-2$, and so Q has distance at most $d'-2+k-m<d$.

Finally, v is incident with the triangle, and v is on the perimeter of the infinite region of N''. So the triangle has distance at most $d' \le d$, by (γ). By (β) and (δ), every other region of M'' also has distance at most d, and

so $\rho(M'') \leq d$, as required.[F1]

We replace every edge of G' which joins two vertices of $V(C_{d'})$ but which is not in $E(C_{d'})$ by two edges in series. The graph thus obtained is a simple planar graph with a minor isomorphic to G, and the drawing we obtain of it has a d'-shell and radius $\leq d$. This completes our inductive argument and hence proves (2.2).

When $r \geq 0$, $s > 0$ are integers, the $r \times s$ *cylinder* is the graph shown in Fig. 1, with r radial lines and s circles.

The "circular" circuits are called the *circles* of the cylinder, and the innermost one is called the *central circle*. When $\lambda \geq 1$ is an integer, the graph Y_λ is a tree as shown in Fig. 2.

Thus Y_λ has $3 \cdot 2^{\lambda-1}$ end-vertices. For $d, \lambda \geq 1$, take the $3 \cdot 2^{\lambda-1} \times d$ cylinder, and let $C_1, \ldots, C_d$ be its circles, in order, where C_d is the central circle. Identify the vertices of the central circle with the end-vertices of Y_λ, in the cyclic order provided by the drawing of Fig. 2. The resulting graph we call $N(d, \lambda)$. (See Fig. 3.) We also set $N(0, \lambda) = Y_\lambda$.

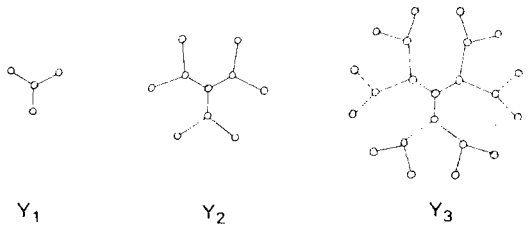
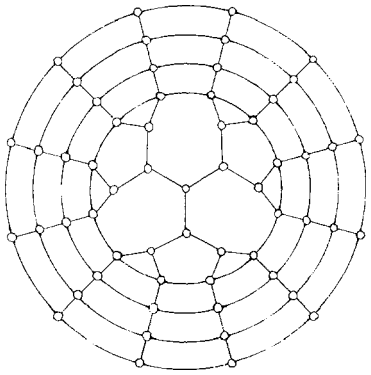


FIGURE 2



[p. 55]

FIGURE 3

[F1] **Missing proof.** The paper leaves the verification of (i)–(iv) for the enlarged graph to the reader. The argument above is the one used in the formalization.

(2.3) *Let G be a simple planar graph, with a drawing M such that $\rho(M) \leq d$ and M has a d-shell $C_1, \ldots, C_d$. Then G is isomorphic to a minor of $N(d, \lambda)$ for some sufficiently large λ.*

Proof. We observe that the graph obtained from G by deleting $V(C_1), \ldots, V(C_d)$ is a forest, since if the corresponding drawing had a finite region R, we would have $d(R) > d$ in M which is impossible. But any forest is isomorphic to a minor of one of Y_1, Y_2, $Y_3, \ldots$. The rest of the proof is obvious, although tedious to write out, and we omit it.

We give the omitted part. If $d = 0$, then G itself is a forest, and so it is isomorphic to a minor of some $Y_\lambda = N(0, \lambda)$. So we assume that $d \geq 1$. Let F be the forest above. By (S1) its vertices are inside C_d, and so all of F is inside C_d, since its edges do not meet C_d.

Step 1: we may assume that F is a tree or is null. Suppose that F has at least two components, let F_1 be one of them, and let F_2 be the union of the others. We claim that the closure of some region of M inside C_d contains a vertex of F_1 and a vertex of F_2. Since F has no circuit and is disjoint from C_d, the only circuit of the drawing formed by F and C_d is C_d. Two points not in a drawing in the plane that are not separated by any circuit of the drawing lie in the same region of it. So the set Ω of points inside C_d that are not in $U(F)$ is a region of this drawing, and in particular it is connected. Every vertex inside C_d is in F, and by (S2) every edge with its interior points inside C_d is in F or joins C_d and F. So Ω is the union of the regions of M inside C_d and the edges of M joining C_d and F. Each such edge is in the closure of the two regions beside it. Let $v_1 \in V(F_1)$ and $v_2 \in V(F_2)$. Since $U(F)$ has no interior point, v_1 is in the closure of Ω, and so in the closure of some region of M inside C_d, and the same holds for v_2. Let Z be the set of regions of M inside C_d whose closure contains a vertex of F_1. If a region in Z has v_2 in its closure, the claim holds. Otherwise Z is non-empty, and so is the set of the other regions of M inside C_d. Suppose that no edge of M in Ω has a region in Z on one side and a region not in Z on the other. Then the regions in Z, with the edges in Ω beside them, form an open subset of Ω, the other regions with the edges beside them form another, and these two sets are disjoint, non-empty and cover Ω, which is impossible since Ω is connected. So some edge g in Ω has a region in Z on one side and a region not in Z on the other. It joins a vertex of C_d and a vertex w of F, and w is in the closure of both regions. Since one of these regions is not in Z, w is not in F_1, and so $w \in V(F_2)$. Thus the region beside g in Z satisfies the claim. Now let q be a region as in the claim, with $u_1 \in V(F_1)$ and $u_2 \in V(F_2)$ in its closure, and add a new edge joining u_1 and u_2, drawn in q. The new graph is simple, since an edge of G joining u_1

and u_2 would be an edge of F joining two of its components. The circuits $C_1, \ldots, C_d$ are still nested, and (S1) and (S2) still hold, since the new edge joins two vertices inside C_d. The new forest F has one component fewer, and G is isomorphic to a minor of the new graph. From now on we use only that G is simple, that $C_1, \ldots, C_d$ is a nested sequence of circuits satisfying (S1) and (S2), and that F is a forest inside C_d, and these properties hold for the new graph. Repeating this step, we may assume that F is a tree or is null.

Step 2: the edges. Every edge of G is in some C_j, or joins C_j and C_{j+1} for some $j < d$, or joins C_d and F, or is in F. For an edge joining two vertices of C_j is in C_j by (S2), and an edge joining C_j and $C_{j'}$ with $j' > j+1$, or C_j and F with $j < d$, would have one end inside C_{j+1} and the other outside, and so it would cross C_{j+1}. For $1 \leq j \leq d$, let G_j be the subgraph of G formed by $C_j, \ldots, C_d$, F, and the edges between them. By (S1), $G_1 = G$.

Step 3: the cyclic orders. We use the following consequence of the Jordan curve theorem.

(A) *Let D' be a closed disc in the interior of a closed disc D in the plane, and let $A_1, \ldots, A_q$ be arcs, each with one end in $bd(D')$, the other end in $bd(D)$, and its other points in the interior of D and not in D', such that no point other than an end is in two of them. Then, going once clockwise around $bd(D')$ from any point that is not an end of one of them, we meet their ends in some order $A_1, \ldots, A_q$, and going once clockwise around $bd(D)$ from a suitable point, we meet their ends in the same order, where arcs with a common end are met together.*

For two of the arcs, together with a portion of $bd(D')$ and a portion of $bd(D)$, bound a closed disc that lies between $bd(D')$ and $bd(D)$, such that both portions run clockwise from one of the two arcs to the other. Every other arc without its ends is connected and misses these two arcs, and so it lies in this disc or outside it. So any three of the arcs meet both circles in the same cyclic order, and (A) follows.

Let $j < d$. Each edge between C_j and C_{j+1} has its interior points between C_j and C_{j+1}, since its interior is connected, misses both circuits, and has an end on each. By (A), applied to the discs bounded by C_{j+1} and C_j, these edges can be listed so that their ends in C_{j+1}, and also their ends in C_j, are in the clockwise cyclic order of these circuits.

Now suppose that F is a tree. Choose a closed disc K inside C_d that contains $U(F)$ in its interior and lies so close to $U(F)$ that it meets no edge of M other than those of F and those joining F and C_d, and meets each edge g joining F and C_d only in points near the end of g in F. (Such discs exist since $U(F)$ is a tree. By a homeomorphism of the plane we may

assume that the edges of F are polygonal, and then the points within a small enough distance of $U(F)$ form one.) Go once clockwise around $bd(K)$, starting from a point near a vertex t of F that is on no edge of M. On the way we pass along the edges of F, each twice, and this gives a closed walk $W = w_0, w_1, \ldots, w_m$ of F with $w_0 = w_m = t$ which passes each edge of F twice. Each k with $0 \leq k \leq m$ is a *visit* of W to w_k, and there $bd(K)$ passes near w_k, through a sector at w_k between two consecutive edges of F (for $k = 0$ and $k = m$, through the parts of such a sector after and before the starting point). An edge g joining w_k and C_d lies near w_k in one of these sectors, and we say that g *leaves* F *at* the corresponding visit. The part of g after its last point in K is an arc from $bd(K)$ to C_d, and its end in $bd(K)$ is where $bd(K)$ passes this visit. By (A), applied to K and the disc bounded by C_d, the edges joining F and C_d can be listed so that the visits at which they leave F are in the order of W, starting from visit 0, and their ends in C_d are in the clockwise cyclic order of C_d.

Step 4: two facts about $N(d', \lambda)$. First, for $d' \geq 1$, $N(d', \lambda)$ is isomorphic to a minor of $N(d', \lambda + 1)$, in such a way that each vertex of a circle is represented by a set containing two consecutive vertices of the corresponding circle. To see this, number the radial lines of $N(d', \lambda + 1)$ as $1, \ldots, 3 \cdot 2^\lambda$ in order, so that for each i the end-vertices of $Y_{\lambda+1}$ on the lines $2i - 1$ and $2i$ have a common neighbour u_i. On each circle, contract the edges between the lines $2i - 1$ and $2i$, and delete the radial edges on the even lines, which are now parallel to those on the odd lines. Then contract the edge joining each u_i to the end-vertex on the line $2i - 1$, and delete the edge joining it to the end-vertex on the line $2i$. Deleting the end-vertices of $Y_{\lambda+1}$ leaves Y_λ, with the vertices u_i as its end-vertices, in the same order. So the result is $N(d', \lambda)$, as required. We call a model of G_j as a minor of $N(d - j + 1, \lambda)$ *cyclic* if each vertex of C_j is represented by a set that contains an interval of the outer circle, and these intervals cover that circle and are in the clockwise cyclic order of C_j. Composing a cyclic model with the minor above doubles the length of each of its intervals, and so the intervals of a cyclic model may be made as long as we wish.

Second, let B_h ($h \geq 1$) be the rooted binary tree in which each vertex at distance less than $h - 1$ from the root has two children, a left one and a right one, and all leaves are at distance $h - 1$ from the root. Its leaves are ordered from left to right. We show by induction on $|E(T)|$ that for every tree T, vertex t of T, and closed walk $w_0, w_1, \ldots, w_m$ of T with $w_0 = w_m = t$ that passes each edge of T twice, there exist $h \geq 1$ and disjoint non-empty sets $X(v) \subseteq V(B_h)$ ($v \in V(T)$), each inducing a connected subgraph, such that the root of B_h is in $X(t)$, for each edge of T with ends u, v some edge of B_h

joins $X(u)$ and $X(v)$, and there are leaves $l_0, \ldots, l_m$ of B_h, in order from left to right, with $l_k \in X(w_k)$ for $0 \le k \le m$. If T has no edge, then $m = 0$, and we take $h = 1$ and $X(t) = \{l_0\}$, where l_0 is the root. Otherwise let e be the first edge of the walk, joining t and $y = w_1$. Deleting e from T leaves two trees, T_y containing y and T_t containing t. Since the walk passes e twice and e is the only edge between T_y and T_t, the walk consists of t, then a closed walk of T_y from y, then a closed walk of T_t from t, and these two walks pass each edge of T_y and of T_t twice. By induction they have sets as above. Sets in $B_{h'}$ give sets in $B_{h'+1}$, by copying them into the subtree at the left child of the root and adding the root to $X(t)$, and so we may assume that both are in $B_{h'}$ for the same h'. Let $h = h' + 2$, let a and b be the children of the root of B_h, and let a_1, a_2 and b_1, b_2 be the children of a and of b, all from left to right. The subtrees at a_2 and b_1 are copies of $B_{h'}$. Copy the sets for T_y into the subtree at a_2 and the sets for T_t into the subtree at b_1, and add to $X(t)$ the root, a, b, and the path from a_1 down to its leftmost leaf, which we take as l_0. Then $X(t)$ induces a connected subgraph, since it contains b_1, and the edge from a to $a_2 \in X(y)$ joins $X(t)$ and $X(y)$. For $l_1, \ldots, l_m$ we take the leaves chosen for the walk of T_y and then those chosen for the walk of T_t. They are in order from left to right, after l_0, since the subtree at a_2 lies to the left of the subtree at b_1. This completes the induction.

Now apply this to F, t and W, and let $\kappa \ge 1$ and $\lambda = h + \kappa$. The subtree of Y_λ at a neighbour of its centre is a copy of B_λ, whose leaves are end-vertices of Y_λ in the order of the central circle, and its vertices at distance at most $h - 1$ from its root form a copy of B_h. Use the sets $X(v)$ in this copy, and add to $X(w_k)$, for $0 \le k \le m$, the descendants of l_k that are not end-vertices of Y_λ. This gives a model of F as a minor of Y_λ with its end-vertices deleted, in which the 2^κ end-vertices below l_k form an interval of the central circle, *owned* by k, whose vertices are adjacent to vertices of $X(w_k)$. These intervals are disjoint and in the order of W around the central circle, and their length 2^κ is as large as we wish.

Step 5: the induction. We show that G_j has a cyclic model for $j = d, d-1, \ldots, 1$. For $j = 1$, since $G_1 = G$, this proves (2.3).

Let $j = d$, let $p = |V(C_d)|$, and let $g_1, \ldots, g_q$ be the edges joining F and C_d, listed as in Step 3, where g_i leaves F at the visit k_i and ends at $x_i \in V(C_d)$. (If F is null, then $q = 0$.) If F is a tree, take the model of F of Step 4 with $2^\kappa \ge q(p+1)$. Since $k_1 \le \cdots \le k_q$, we can choose for each i an end-vertex z_i in the interval owned by k_i, so that $z_1, \ldots, z_q$ are in order around the central circle, with at least p other vertices of this circle between any two of them that are consecutive in the cyclic order. Since $x_1, \ldots, x_q$ are in the cyclic order of C_d, we can then divide the central circle

into intervals $I(x)$ ($x \in V(C_d)$), each with at least one vertex and in the clockwise cyclic order of C_d, with $z_i \in I(x_i)$ for each i. (The spare vertices between consecutive z_i leave room for the intervals of the vertices of C_d that lie strictly between x_i and x_{i+1} in the cyclic order.) If $q = 0$, we only need $3 \cdot 2^{\lambda-1} \geq p$. Represent each $x \in V(C_d)$ by $I(x)$, each edge of C_d by the edge of the central circle between the intervals of its ends, which are consecutive since C_d is a circuit, each vertex and edge of F as in Step 4, and each g_i by the edge of Y_λ joining z_i to its neighbour, which is in the set of the end w_{k_i} of g_i in F. This is a cyclic model of G_d in $N(1, \lambda)$.

Now let $j < d$, let $p = |V(C_j)|$, and let $g_1, \ldots, g_q$ be the edges between C_j and C_{j+1}, listed as in Step 3, where g_i joins $y_i \in V(C_j)$ and $x_i \in V(C_{j+1})$. Take a cyclic model of G_{j+1} in $N(d-j, \lambda)$ whose intervals each have at least $q(p+1)$ vertices, which exists by the first fact of Step 4. Let $I(x)$ be its interval for $x \in V(C_{j+1})$. The graph $N(d-j+1, \lambda)$ consists of a copy of $N(d-j, \lambda)$, a new outer circle, and a radial edge joining each vertex of the new circle to the vertex of the old outer circle on the same radial line. Use the cyclic model of G_{j+1} in the copy. Since $x_1, \ldots, x_q$ are in the cyclic order of C_{j+1} and the intervals are long, we can choose radial lines $\ell_1, \ldots, \ell_q$, in order around the circles, such that ℓ_i meets the old outer circle in $I(x_i)$, with at least p other radial lines between any two of them that are consecutive in the cyclic order. Since $y_1, \ldots, y_q$ are in the cyclic order of C_j, we can then divide the new outer circle into intervals $I'(y)$ ($y \in V(C_j)$), each with at least one vertex and in the clockwise cyclic order of C_j, such that ℓ_i meets it in $I'(y_i)$, as in the case $j = d$. Represent each $y \in V(C_j)$ by $I'(y)$, each edge of C_j by the edge of the new circle between the intervals of its ends, and each g_i by the radial edge of ℓ_i joining the two outer circles. With the model of G_{j+1}, this is a cyclic model of G_j in $N(d-j+1, \lambda)$, by Step 2. This completes the induction and the proof of (2.3).[F2]

(2.4) *If G is a planar graph with radius $\leq d$ then G is isomorphic to a minor of $N(d, \lambda)$ for some sufficiently large λ.*

This follows from (2.1), (2.2), and (2.3).

(2.5) *For all $d \geq 0$, $\lambda \geq 1$, the graph $N(d, \lambda)$ has tree-width $\leq 3d + 1$.*

Proof. If $d = 0$ the result is clear, and we assume $d > 0$. We construct a tree-decomposition of $N(d, \lambda)$ as follows. Let T be the tree obtained from $N(d, \lambda)$ by deleting the $3 \cdot 2^{\lambda-1} \times d$ cylinder, and let C be the central circle of the cylinder. For each vertex v of C, let P_v be the set of all vertices of the cylinder on the same radial line as v. Thus $|P_v| = d$. Let t be a vertex

[F2] **Missing proof.** The paper omits this part of the proof. The argument above is the one used in the formalization.

of T. Now t is on precisely three regions R_1, R_2, R_3 say (taking a drawing M of $N(d,\lambda)$ as in Fig. 3). For $j = 1,2,3$, there are two vertices of C on R_j and they are adjacent. Let them be u_j, v_j, where v_j follows u_j in the anticlockwise orientation of C. Define

$$X_t = P_{v_1} \cup P_{v_2} \cup P_{v_3} \cup \{t, t'\},$$

where $t' = t$ if t is the "center" of T, and t' is the neighbour of t closer to the center if ~~t'~~ t is not the center of T.[F3] It is easy to verify that $(T, \mathcal{X})$ is a tree- [p. 56] decomposition of $N(d,\lambda)$ of width $\leq 3d+1$, where $\mathcal{X} = (X_t : t \in V(T))$. This completes the proof.

We give the details. If $d = 0$, then $N(0,\lambda) = Y_\lambda$ is a tree, and the sets $\{t, t'\}$ $(t \in V(Y_\lambda))$, where t' is the neighbour of t closer to the center (and $t' = t$ for the center), form a tree-decomposition of width ≤ 1 over the tree Y_λ. Now let $d > 0$. Then $|X_t| \leq 3d+2$, and so the width is at most $3d+1$. For each edge of C with ends u, v, where v follows u, let $R(u,v)$ be the region of M inside C incident with this edge. It is bounded by the edge and the path of Y_λ between u and v, and the vertices of T on $R(u,v)$ are the internal vertices of this path, of which there is at least one. Since the sets P_v are disjoint, a vertex x of the cylinder with $x \in P_v$ is in X_t if and only if t is on $R(u,v)$. So the set of all t with $x \in X_t$ is non-empty, and consists of the internal vertices of the path of Y_λ between u and v. For each of them except the one nearest to the center, the neighbour t' closer to the center is also on this path. So this set is the vertex set of a subtree of T. A vertex s of T is in X_t if and only if $t = s$ or $t' = s$, and so the set of these t is the vertex set of a subtree of T. This proves (W1) and (W3). For (W2), an edge of T joins some $t \neq t'$ to t', and both are in X_t. An edge of Y_λ joining an end-vertex v to t has t on $R(u,v)$, and so $P_v \cup \{t\} \subseteq X_t$. An edge on a radial line has both ends in some P_v. Finally, an edge of a circle joins P_u and P_v for some edge of C with ends u, v, where v follows u. Let w be the vertex of C that u follows, and let s be the neighbour of u in Y_λ. Then s is on $R(w,u)$ and on $R(u,v)$, and so $P_u \cup P_v \subseteq X_s$.[F4]

Finally, we need

(2.6) *If G' is isomorphic to a minor of G then the tree-width of G' is not greater than the tree-width of G.*

The proof is clear. To see this, let $(T, \mathcal{X})$ be a tree-decomposition of G of width w, and let G' be a minor of G. Then there are disjoint non-empty sets

[F3] **Misprint.** The vertex t' is being defined here, and the condition is on t.

[F4] **Missing proof.** The paper leaves this verification to the reader. The argument above is the one used in the formalization.

$Y(v) \subseteq V(G)$ ($v \in V(G')$), each inducing a connected subgraph of G, and for each edge e of G' an edge $f(e)$ of G, such that if e has ends u, v, then $f(e)$ has one end in $Y(u)$ and the other in $Y(v)$. (Take for $Y(v)$ the set of vertices of G that are contracted into v, and for $f(e)$ the edge of G that becomes e.) For $t \in V(T)$ let X'_t be the set of all $v \in V(G')$ with $Y(v) \cap X_t \neq \emptyset$. Choosing for each $v \in X'_t$ a vertex in $Y(v) \cap X_t$ gives distinct vertices for distinct v, since the sets $Y(v)$ are disjoint, and so $|X'_t| \leq |X_t|$. We claim that $(T, (X'_t : t \in V(T)))$ is a tree-decomposition of G', which therefore has width $\leq w$. For (W1), each $Y(v)$ is non-empty, and every vertex of G is in some X_t. For (W2), let e be an edge of G' with ends u, v. Then $f(e)$ has ends $a \in Y(u)$ and $b \in Y(v)$, and $a, b \in X_t$ for some t, by (W2) for G. Then $u, v \in X'_t$. For (W3), let t' be on the path of T between t_1 and t_2, and let $v \in X'_{t_1} \cap X'_{t_2}$. Suppose that $Y(v) \cap X_{t'} = \emptyset$. Choose $y_1 \in Y(v) \cap X_{t_1}$ and $y_2 \in Y(v) \cap X_{t_2}$. Then $t_1, t_2 \neq t'$. We use (3.1) and (3.2) below, which are proved there from (W1)–(W3) alone. Since $Y(v)$ induces a connected subgraph of G with no vertex in $X_{t'}$, applying (3.2) along a path of $G|Y(v)$ from y_1 to y_2 shows that $T_{t'}(y_1) = T_{t'}(y_2)$. By (3.1) this branch of T at t' contains t_1 and t_2. But t_1 and t_2 are in different branches of T at t', since t' is on the path of T between them. This is a contradiction, and so $Y(v) \cap X_{t'} \neq \emptyset$, that is, $v \in X'_{t'}$.[F5] From (2.4), (2.5), and (2.6) we deduce

(2.7) *If G is planar and has radius $\leq d$ then its tree-width is at most $3d + 1$.*

This completes the proof of the main result of this section. We shall also need the following lemma, which is easy ~~and is left to the reader~~.

(2.8) *Let G be a planar graph with radius $\geq d$, and let M be a drawing of G. Then there is a nested sequence $C_1, C_2, \ldots, C_d$ of circuits of M.*

Proof. We may assume that $d \geq 1$. Since G has radius $\geq d$, we have $\rho(M) \geq d$, and so there is a region R^* of M with $d(R^*) \geq d$. Choose a point p in the infinite region of M and a point a in R^*. For $0 \leq t \leq d - 1$, let U_t be the union of the closures of all regions R of M with $d(R) \leq t$. Then U_t is connected, since each such region is joined to the infinite region by a sequence of regions as in the definition of $d(R)$, and consecutive regions of the sequence have a common vertex. Moreover $a \notin U_t$, since R^* is open and meets no other region. Let B_t be the subgraph of G consisting of the vertices and edges that are incident both with a region R with $d(R) \leq t$ and with a region R' with $d(R') > t$. If a vertex is incident with a region R, then every region incident with it has distance at most $d(R) + 1$.

[F5] **Missing proof.** The paper says that the proof is clear. The argument above is the one used in the formalization.

(1) *For $0 \le t < t' \le d-1$, B_t is contained in U_t, and no point of $B_{t'}$ is in U_t.*

The first statement holds by the definition of B_t. For the second, a vertex of $B_{t'}$ in U_t would be incident with a region of distance $\le t$ and with a region of distance $> t' \ge t+1$, which is impossible. If some other point of an edge f of $B_{t'}$ is in U_t, then f is incident with a region of distance $\le t$, and so are its ends. But its ends are also incident with a region of distance $> t'$, since f is, and this is impossible as before. This proves (1).

(2) *For $0 \le t \le d-1$ there is a circuit C_{t+1} of B_t with R^* inside and the infinite region of M outside.*

Suppose not. Then for every circuit of B_t, one of the two closed discs in the sphere bounded by it contains neither p nor a. It follows that p and a are in the same region of the drawing of B_t, since two points in different regions of a drawing are separated by some circuit of it. So there is a curve from p to a that meets no point of B_t. It starts in U_t and ends outside U_t, and so it passes through a point of the boundary of U_t. This point is in the closure of a region of distance $\le t$ and in the closure of a region of distance $> t$, and so it is a vertex or a point of an edge of B_t, which is impossible. So some circuit of B_t is such that each of its two closed discs contains p or a. Since p, a are not on the circuit, one disc contains p and not a, and the other contains a and not p. This proves (2).

Now let $0 \le t < t' \le d-1$. By (1), C_{t+1} is contained in U_t, and $C_{t'+1}$ is disjoint from U_t. So C_{t+1} and $C_{t'+1}$ have no common vertex. Moreover U_t is connected, contains the infinite region and is disjoint from $C_{t'+1}$, and so it is outside $C_{t'+1}$. In particular C_{t+1} is outside $C_{t'+1}$. So the part of the sphere inside $C_{t'+1}$ is connected, contains R^*, and is disjoint from C_{t+1}, and therefore it is inside C_{t+1}. Hence $C_{t'+1}$, which is in its closure and has no common vertex with C_{t+1}, is inside C_{t+1}. Thus $C_1, \ldots, C_d$ is a nested sequence of circuits of M.[F6]

3. Sleeve Unions

We shall require some results about tree-decompositions which are proved in [2], but which we prove again here for the reader's convenience.

Let $(T, \mathcal{X})$ be a tree-decomposition of G, where $\mathcal{X} = (X_t : t \in V(T))$. For each $t \in V(T)$, the connected components of $T \setminus t$ are called the *branches* of T at t.

[F6] **Missing proof.** The paper says that (2.8) is easy and leaves it to the reader. The argument above is the one used in the formalization, where U_t is built from the layers of regions at distance at most t.

(3.1) *For $t \in V(T)$ and $v \in V(G)$, either $v \in X_t$, or there is a branch of T at t which contains all $t' \in V(T)$ with $v \in X_{t'}$.*

This follows immediately from (W1), (W3). If $v \notin X_t$, let $T_t(v)$ denote this branch. (It is unique.)

(3.2) *If $v, v' \notin X_t$ and v, v' are adjacent in G then $T_t(v) = T_t(v')$.*

Proof. By (W2), there exists $t' \in V(T)$ such that $v, v' \in X_{t'}$. Then $t' \neq t$, and so t' is in some branch B of T at t. But then $B = T_t(v)$, because $v \in X_{t'}$, and similarly $B = T_t(v')$.

(3.3) *If $v, v' \notin X_t$ and v, v' are not separated in G by X_t, then $T_t(v) = T_t(v')$.*

Proof. Let $v_1, v_2, \ldots, v_k$ be a sequence of vertices of G not in X_t, each adjacent to the next, with $v_1 = v$ and $v_k = v'$. Then by (3.2),

$$T_t(v_1) = T_t(v_2) = \cdots = T_t(v_k)$$

and so $T_t(v) = T_t(v')$.

[p. 57] (3.4) *Let e be an edge of T with ends t, t' say, and let N, N' be the vertex sets of the two components of $T \setminus e$. Then $X_t \cap X_{t'}$ separates $\bigcup(X_n : n \in N)$ and $\bigcup(X_n : n \in N')$.*

Proof. Suppose not. Then from (W1), there exist

$$v, v' \in V(G) - (X_t \cap X_{t'})$$

with $v \in X_n$ and $v' \in X_{n'}$ say, where $n \in N$ and $n' \in N'$, such that either $v = v'$ or v, v' are adjacent in G. From (W1), (W2) there exists $t'' \in V(T)$ such that $v, v' \in X_{t''}$. We assume without loss of generality that $t'' \in N'$. Then t, t' are both on the path of T between n and t'', and so $X_n \cap X_{t''} \subseteq X_t, X_{t'}$ by (W3). Hence $v \in X_t \cap X_{t'}$, a contradiction, as required.

(3.5) *Let $|V(T)| \geq 2$, and for each $t \in V(T)$ let G_t be a connected subgraph of G with $V(G_t) \cap X_t = \emptyset$. Then there exist $t, t' \in V(T)$, adjacent in T, such that $X_t \cap X_{t'}$ separates $V(G_t)$ and $V(G_{t'})$ in G.*

Proof. For each $t \in V(T)$, let B_t be a branch of T at t such that $T_t(v) = B_t$ for all $v \in V(G_t)$. This is possible by (3.3). (We use the fact that $|V(T)| \geq 2$ to ensure that T has a branch at t.) Let e_t be the edge of T joining t to a vertex of B_t. Now T has fewer edges than vertices, and so there exist distinct $t, t' \in V(T)$ with $e_t = e_{t'}$. Then t, t' are adjacent in T. Let N, N' be the vertex sets of the two components of $T \setminus e_t$, with $t \in N, t' \in N'$. Then $N = V(B_{t'}), N' = V(B_t)$, and

$$V(G_t) \subseteq \bigcup(X_n : n \in N'), \qquad V(G_{t'}) \subseteq \bigcup(X_n : n \in N)$$

and the result follows from (3.4).

Our object in this section is to introduce the "sleeve union" of two graphs, and to prove that the tree-width of the union is the maximum of the tree-widths of the two pieces. We need some further lemmas.

Let S be the $r \times s$ cylinder, where $r \leq 2s$, and let $C_1, \ldots, C_s$ be its circles in order, where C_s is the central circle.

(3.6) *Suppose $X \subseteq V(S)$ and $|X \cap V(C_j)| \geq 2$ $(1 \leq j \leq s)$. Then there are r vertex-disjoint paths of S, each between X and $V(C_s)$.*

Proof. By Menger's theorem, it is enough to show that if $Y \subseteq V(S)$ and $|Y| < r$, there is a path of S between X and $V(C_s)$ which avoids Y. Thus, let $Y \subseteq V(S)$, with $|Y| < r$. Now since $r \leq 2s$, there exists j with $1 \leq j \leq s$ such that $|V(C_j) \cap Y| \leq 1$. Choose $v \in V(C_j)$ such that $V(C_j) \cap Y \subseteq \{v\}$. Then $V(C_j) - \{v\}$ contains no vertex of Y. However, it does contain a vertex of X, since $|V(C_j) \cap X| \geq 2$. Let L be the set of vertices in a radial line of S such that $L \cap Y = \emptyset$. (This is possible since S has r radial lines and $|Y| < r$.)

[p. 58] Then $L \cup (V(C_j) - \{v\})$ is a subset of $V(S) - Y$, and induces a subgraph of S which contains a path of S between X and C_s avoiding Y, as required.

(3.7) *Let $(T, \mathcal{X})$ be a tree-decomposition of S, where $\mathcal{X} = (X_t : t \in V(T))$. Then there exist $t_0 \in V(T)$ and r vertex-disjoint paths of S, each between X_{t_0} and C_s.*

Proof. Let the vertex sets of the radial lines of S be $L_1, \ldots, L_r$. If some $t \in V(T)$ has $X_t \cap L_i \neq \emptyset$ for all i $(1 \leq i \leq r)$, the theorem is obviously true. We assume then that for each $t \in V(T)$ there exists i with $1 \leq i \leq r$ such that $X_t \cap L_i = \emptyset$. From (3.5) there exist $t, t' \in V(T)$, adjacent in T, such that $X_t \cap X_{t'}$ separates L_i and $L_{i'}$ for some i, i' with $1 \leq i, i' \leq r$, and $X_t \cap X_{t'}$ intersects neither of $L_i, L_{i'}$. Clearly $i \neq i'$, and for $1 \leq j \leq k$ the two paths in C_j from L_i to $L_{i'}$ both meet $X_t \cap X_{t'}$, and so $|X_t \cap V(C_j)| \geq 2$ $(1 \leq j \leq s)$. The result follows from (3.6).

(3.8) *With $S, C_1, \ldots, C_s$ as before, let $V(C_s) = \{u_1, \ldots, u_r\}$. Let H be another graph, vertex-disjoint from S, and let $v_1, \ldots, v_r$ be distinct vertices of H. Construct G by making the identifications $u_1 = v_1, \ldots, u_r = v_r$. If G has tree-width $\leq w$, then H has a tree-decomposition $(T, \mathcal{X})$ where $\mathcal{X} = (X_t : t \in V(T))$ say, of width $\leq w$, such that $\{v_1, \ldots, v_r\} \subseteq X_{t_0}$ for some $t_0 \in V(T)$.*

Proof. Let $(T, \mathcal{X}_1)$ be a tree-decomposition of G of width $\leq w$, where

$\mathcal{X}_1 = (Y_t : t \in V(T))$ say. Define

$$\mathcal{X}_2 = (Y_t \cap V(S) : t \in V(T)).$$

Clearly $(T, \mathcal{X}_2)$ is a tree-decomposition of S, and so by (3.7) there exists $t_0 \in V(T)$ and vertex-disjoint paths $P_1, \ldots, P_r$ of S, each between $Y_{t_0} \cap V(S)$ and $\{u_1, \ldots, u_r\}$. Order these so that for $1 \le i \le r$, u_i is the terminal vertex of P_i. For $t \in V(T)$, define

$$X_t = (Y_t \cap V(H)) \cup \{v_i : 1 \le i \le r, Y_t \cap V(P_i) \neq \emptyset\}.$$

Put $\mathcal{X} = (X_t : t \in V(T))$. We claim that $(T, \mathcal{X})$ is a tree-decomposition of H. For suppose that $t, t', t'' \in V(T)$, and t' lies on the path of T between t and t'', and suppose that $v \in X_t, X_{t''}$. We must show that $v \in X_{t'}$. Now if $v \neq v_1, \ldots, v_r$, then $v \in Y_t \cap Y_{t''} \subseteq Y_{t'}$, and so $v \in X_{t'}$. If $v = v_i$ say, then $Y_t \cap V(P_i) \neq \emptyset$ and $Y_{t''} \cap V(P_i) \neq \emptyset$. But $Y_{t'}$ separates Y_t and $Y_{t''}$ in G by (3.3), and so $Y_{t'} \cap V(P_i) \neq \emptyset$ and again $v \in X_{t'}$. Thus $(T, \mathcal{X})$ is a tree-decomposition of H. Clearly it has width $\le w$, and

$$\{v_1, \ldots, v_r\} \subseteq X_{t_0},$$

as required.

[p. 59] If H is a minor of G then each vertex u of H is formed by taking a nonempty subset $Z_u \subseteq V(G)$ and identifying the vertices in Z_u by contraction. If $v \in Z_u$ we write $v \to u$.

A *separation* of a graph G is a pair (H_1, H_2), where H_1, H_2 are subgraphs of G such that

$$H_1 \cup H_2 = G, \qquad E(H_1) \cap E(H_2) = \emptyset.$$

Let (H_1, H_2) be a separation of G. Let

$$V(H_1) \cap V(H_2) = \{v_1, \ldots, v_r\},$$

say. Let s be the least positive integer such that $2s \ge r$.[F7] Let S_1, S_2 be graphs, both isomorphic to the $r \times s$ cylinder, and let the vertex set of the central circles of S_1, S_2 be $\{u_1^1, \ldots, u_r^1\}$ and $\{u_1^2, \ldots, u_r^2\}$, respectively, in order. Suppose that S_1 is a proper minor of H_2, where $v_1 \to u_1^1, \ldots, v_r \to u_r^1$, and S_2 is a proper minor of H_1 where $v_1 \to u_1^2, \ldots, v_r \to u_r^2$.

[F7] **Degenerate case.** For $r = 0$ the least integer s with $2s \ge r$ is $s = 0$, but the $r \times s$ cylinder is defined only for $s > 0$ (page 54). The change affects only $r = 0$, where s becomes 1. With this s, (3.9) holds for all $r \ge 0$.

Let G_k be the graph obtained from H_k, S_k by making the identifications $v_i = u_i^k$ $(1 \le i \le r)$ $(k = 1, 2)$. We say that G is the *sleeve union* of G_1, G_2.

(3.9) *If G is the sleeve union of G_1, G_2 then G_1, G_2 are both isomorphic to proper minors of G.*

This is clear from the construction.

(3.10) *If G is the sleeve union of G_1, G_2 then the tree-width of G is the maximum of the tree-widths of G_1, G_2.*

Proof. Let w be the maximum of the tree-widths of G_1, G_2. By (3.9) and (2.6), the tree-width of G is at least w. We must prove the reverse inequality. G_1 has tree-width $\le w$, and so by (3.8), H_1 has a tree-decomposition $(T_1, \mathcal{X}_1)$ of width $\le w$, where $\mathcal{X}_1 = (X_t : t \in V(T_1))$ say, such that $\{v_1, \ldots, v_r\} \subseteq X_{t_1}$ for some $t_1 \in V(T_1)$. Define $T_2, \mathcal{X}_2, t_2$ similarly for H_2, arranged so that T_1, T_2 are disjoint. Let T be the tree constructed from T_1, T_2 by adding an edge joining t_1, t_2. Put $\mathcal{X} = (X_t : t \in V(T))$. It is easy to see that $(T, \mathcal{X})$ is a tree-decomposition for G of width $\le w$, as required.

4. The Production of Cylinders

Let $r \ge 0$, $s \ge 1$ be integers, and let M be a drawing of a planar graph G. We say that M *majors* the $r \times s$ cylinder if there is a nested sequence $C_1, \ldots, C_s$ of circuits of M and vertex-disjoint paths $P_1, \ldots, P_r$ of G, each between $V(C_1)$ and $V(C_s)$, such that for $1 \le i \le r$ and $1 \le j \le s$, $P_i \cap C_j$ is a path.

[p. 60] (4.1) *Let $r \ge 0$, $s \ge 1$ be integers, and let M be a drawing of the planar graph G. Suppose that there is a nested sequence $C_1, \ldots, C_s$ of circuits of M such that there are r vertex-disjoint paths of G, each between $V(C_1)$ and $V(C_s)$. Then M majors the $r \times s$ cylinder.*

Proof. Let $C_1, \ldots, C_s$ be a nested sequence of circuits of M, and let $P_1, \ldots, P_r$ be vertex-disjoint paths of G, each between $V(C_1)$ and $V(C_s)$. Choose $C_1, \ldots, C_s, P_1, \ldots, P_r$ with

$$|E(C_1 \cup \cdots \cup C_s \cup P_1 \cup \cdots \cup P_r)| \text{ minimum.} \tag{1}$$

We shall show that for $1 \le i \le r$ and $1 \le j \le s$, $P_i \cap C_j$ is a path.

Suppose that $C_1 \cap P_i$ is not a path for some i, say $i = 1$. Let e be the first edge of P_1 not in C_1. Then there is a subpath of P_1 (P_1' say) between C_1 and C_s not using e; and then e lies in none of $C_1, \ldots, C_s, P_1', P_2, \ldots, P_r$, contrary to (1). Thus $C_1 \cap P_i$ is a path, and similarly $C_s \cap P_i$ is a path, for each i $(1 \le i \le r)$.

Suppose for a contradiction that there is some value of j with $1 \le j \le s$ such that $P_i \cap C_j$ is not a path for some i $(1 \le i \le r)$. Choose such a value of j, as small as possible. Then by the foregoing, $2 \le j \le s-1$.

Suppose that for some i $(1 \le i \le r)$ there are two distinct vertices v, v' say of $P_i \cap C_j$, such that all edges and all interior vertices of the subpath of P_i between v and v' are outside C_j. Let this subpath be P, say. Now no interior vertex of P is in $V(C_{j'})$ for any j' with $1 \le j' \le s$; for all are outside C_j, and none are on C_{j-1}, by the minimality of j, as is easily seen. Let the subpaths of C_j between v and v' be Q_1, Q_2. Then $P \cup Q_1, P \cup Q_2$ are circuits, vertex-disjoint from every $C_{j'}$ $(1 \le j' \le s, j' \ne j)$. Now C_{j+1} is inside one of them, say $P \cup Q_1$. But then

$$C_1, \ldots, C_{j-1}, P \cup Q_1, C_{j+1}, \ldots, C_s$$

is a nested sequence of circuits, contrary to (1), for some edge of Q_2 is not in any of $P_1, \ldots, P_r$. Thus there is no such value of i.

It follows that

> for every vertex v of C_j, if v is on P_i where $1 \le i \le r$ then there is a subpath of P_i between v and C_s which uses no edges outside C_j. (2)

By the choice of j there exists some i $(1 \le i \le r)$ such that $P_i \cap C_j$ is not a path; and then there are distinct vertices v, v' of $P_i \cap C_j$ such that all edges and all interior vertices of the subpath of P_i between v and v' are inside C_j. Let this subpath be P, and let Q_1, Q_2 be the paths of C_j between v and v'. Now $P_i \cap C_s$ is a path, and so P does not meet C_s. Clearly there is no path of G between $V(Q_1)$ and $V(Q_2)$ which avoids $V(P)$ and which has no edges outside C_j. From these two facts it follows that for one of Q_1, Q_2, say Q_k, [p. 61] there is no path of G from $V(Q_k)$ to C_s which avoids $V(P)$ and has no edges outside C_j. But then by (2), no vertex of Q_k is on any $P_{i'}$ with $1 \le i' \le r$, $i' \ne i$. Let P' be the subgraph of G obtained from $P_i \cup Q_k$ by deleting the edges and interior vertices of P. Then P' contains a path between C_1 and C_s, P_i' say, and

$$P_1, \ldots, P_{i-1}, P_i', P_{i+1}, \ldots, P_r$$

are vertex-disjoint. But this contradicts (1), since the first edge of P is not in any of $C_1, \ldots, C_s$.

Thus the choice of j is impossible, and the theorem is proved.

(4.2) *Let $r \ge 0$, $s \ge 1$ be integers, and let M be a drawing of the planar graph G. Let $C_1, \ldots, C_s$ be a nested sequence of circuits of M. Then one of the following holds:*

(i) *M majors the $r \times s$ cylinder,*

(ii) *there exists $X \subseteq V(G)$ with $|X| < r$, such that X separates $V(C_1)$ and $V(C_s)$.*

Proof. If (ii) is false, then by Menger's theorem there are r vertex-disjoint paths of G between $V(C_1)$ and $V(C_s)$, and so (i) holds, from (4.1).

(4.3) *Let $r \geq 0$ be an integer, and let $r' = r + 1$ if r is odd, and $r' = r$ if r is even. Let $s > r'$ be an integer. Let M be a drawing of the planar graph G, and suppose that M majors the $r \times s$ cylinder. Then either M majors the $(r+1) \times (s - r')$ cylinder, or G is expressible as the sleeve union of two planar graphs.*

Proof. Let $C_1, \ldots, C_s$ be a nested sequence of circuits of M, and let $P_1, \ldots, P_r$ be vertex-disjoint paths of G, each between $V(C_1)$ and $V(C_s)$, such that for $1 \leq i \leq r$ and $1 \leq j \leq s$, $P_i \cap C_j$ is a path. Put $r' = 2d$. Now $C_{d+1}, C_{d+2}, \ldots, C_{s-d}$ is a nested sequence of $s - r'$ circuits of M. By (4.2) either M majors the $(r+1) \times (s-r')$ cylinder or there exists $X \subseteq V(G)$ with $|X| \leq r$ which separates $V(C_{d+1})$ and $V(C_{s-d})$. We assume the latter. For $1 \leq i \leq r$, let P_i' be the subpath of P_i between $V(C_{d+1})$ and $V(C_{s-d})$, with no internal vertex in $V(C_{d+1})$ or $V(C_{s-d})$. Then $X \cap V(P_i') \neq \emptyset$; choose $v_i \in X \cap V(P_i')$ $(1 \leq i \leq r)$. We have $\{v_1, \ldots, v_r\} \subseteq X$ and $|X| \leq r$, and so $X = \{v_1, \ldots, v_r\}$. Let (H_1, H_2) be a separation of G with $V(H_1) \cap V(H_2) = X$, where $C_1, \ldots, C_{d+1}$ are circuits of H_1 and $C_{s-d}, \ldots, C_s$ are circuits of H_2. Now H_1 has a proper minor isomorphic to the $r \times d$ cylinder in the required way for sleeve unions, and so has H_2; thus G is expressible as a sleeve union of two planar graphs. This completes the proof.

(4.4) *Let $r \geq 0$, $s \geq 1$ be integers, and let M be a drawing of the planar graph G. Suppose that G is not expressible as a sleeve union of two planar* [p. 62] *graphs, and M does not major the $r \times s$ cylinder. Then G has radius $< k$, where*

$$\begin{aligned} k &= s + \frac{1}{2}r^2 && (r \text{ even}) \\ &= s + \frac{1}{2}(r^2 - 1) && (r \text{ odd}). \end{aligned}$$

Proof. We proceed by induction on r. If $r = 0$ the result is true by (2.8). We assume then that $r > 0$. If r is even, then by (4.3) M does not major the $(r-1) \times (s+r)$ cylinder, and hence by induction the radius of G is

$$< s + r + \frac{1}{2}((r-1)^2 - 1) = s + \frac{1}{2}r^2$$

as required. If r is odd we argue similarly.

(4.5) *Let $r \geq 0$, $s \geq 1$ be integers. If G is a planar graph which is not expressible as the sleeve union of two planar graphs, then either G has a minor isomorphic to the $r \times s$ cylinder, or it has tree-width at most*

$$3(s + \frac{1}{2}r^2) - 2 \qquad (r \text{ even})$$
$$3(s + \frac{1}{2}(r^2 - 1)) - 2 \qquad (r \text{ odd}).$$

Proof. Take a drawing M of G. If M majors the $r \times s$ cylinder then G has a minor isomorphic to the $r \times s$ cylinder. If not, then G has radius $< k$, where

$$k = s + \frac{1}{2}r^2 \qquad (r \text{ even})$$
$$= s + \frac{1}{2}(r^2 - 1) \qquad (r \text{ odd})$$

by (4.4), and the result follows from (2.7).

5. The Main Result

Now we prove (1.4). We need two more lemmas.

(5.1) *For every planar graph H there is a number $r \geq 1$ such that the $r \times r$ cylinder has a minor isomorphic to H.*

The reader will probably have no trouble convincing himself of this. A proof of (5.1) and of a stronger result will appear in [5].

(5.2) *Let $r \geq 1$ be an integer. If G is planar and has no minor isomorphic to the $r \times r$ cylinder then G has tree-width $\leq \frac{3}{2}(r^2 + 2r) - 2$.*

Proof. We proceed by induction on the size of G (that is, $|V(G)| +$ [p. 63] $|E(G)|$). Suppose that G is expressible as the sleeve union of two planar graphs G_1, G_2. Then G_1, G_2 are isomorphic to proper minors of G, by (3.9), and so each has no minor isomorphic to the $r \times r$ cylinder. Moreover, the sizes of G_1, G_2 are less than the size of G, and so by induction G_1, G_2 both have tree-width $\leq \frac{3}{2}(r^2 + 2r) - 2$. Hence by (3.10), G has tree-width $\leq \frac{3}{2}(r^2 + 2r) - 2$. We may therefore assume that G is not expressible as a sleeve union of two planar graphs; but then the result follows from (4.5). This completes the proof.

We deduce (1.4), which we restate.

(5.3) *For every planar graph H there is a number w such that every planar graph with no minor isomorphic to H has tree-width $\leq w$.*

Proof. From (5.1) there is a number $r \geq 1$ such that the $r \times r$ cylinder has a minor isomorphic to H. Let G be any planar graph with no minor

isomorphic to H. Then G certainly has no minor isomorphic to the $r \times r$ cylinder, and so by (5.2) G has tree-width $\leq \frac{3}{2}(r^2 + 2r) - 2$. Thus (5.3) is true with any number $w \geq \frac{3}{2}(r^2 + 2r) - 2$.

6. Remarks

One natural question concerning the tree-width of planar graphs is, what is the tree-width of the geometric dual? It seems that the tree-width of a planar graph and the tree-width of its geometric dual are approximately equal—indeed, we have convinced ourselves that they differ by at most one. However, they need not be equal. The two graphs of Fig. 4 are duals; but the first has tree-width 4 while the second has tree-width 5.

Concerning our main theorem (1.4), one method of generalization is the

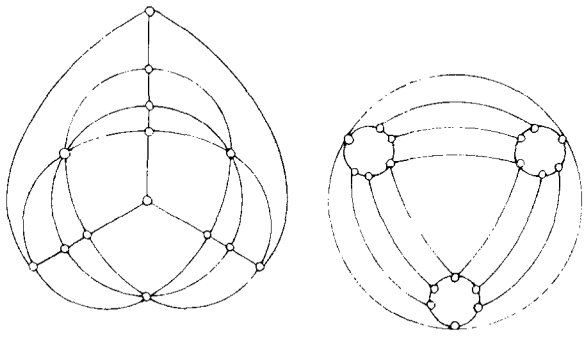


Figure 4

[p. 64] result of [4] which was described after the proof of (1.5). There is another, however, concerning surfaces of higher genus, which rather surprisingly is simpler than (1.4) itself. A proof of the following will appear in [6].

(6.1) *Let S be a surface of genus $g > 0$, and let H be a graph which may be embedded in S. Then there is a number k with the following property. For every graph G which has no minor isomorphic to H but which can be embedded in S, there exists $X \subseteq V(G)$ with $|X| \leq k$ such that $G \setminus X$ can be embedded on a surface of genus $< g$.*

References

1. N. ROBERTSON AND P. D. SEYMOUR, Graph minors. I. Excluding a forest, *J. Combin. Theory Ser. B* **35** (1983), 39–61.

2. N. ROBERTSON AND P. D. SEYMOUR, Graph minors. II. Algorithmic aspects of tree-width, submitted.

3. N. ROBERTSON AND P. D. SEYMOUR, Graph minors. IV. Tree-width and well-quasi-ordering, submitted.

4. N. ROBERTSON AND P. D. SEYMOUR, Graph minors. V. Excluding a planar graph, submitted.

5. N. ROBERTSON AND P. D. SEYMOUR, Graph minors. VI. Disjoint paths on a surface, in preparation.

6. N. ROBERTSON AND P. D. SEYMOUR, Graph minors. VII. Disjoint paths on a surface, in preparation.